VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi -590018

An Internship Report

On

Data Analytics using Python

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering

In

Artificial Intelligence & Data Science

Submitted by

SAMITHA HALLI

1ST22AD043

Internship work carried out at

Dyashin Technosoft Pvt Ltd

#32/33/5B, 1st Floor, Annapurna

Compound Block II,

Kanakapura Road, J P Nagar,

Bengaluru, Karnataka – 560078,India.

Under the Internal Guidance of

Dr. MAMATHA M

Associate Professor

Dept. of AI&DS

SaIT Bengaluru

SAMBHRAM INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

MS PALYA, BENGALURU – 560097

2025-2026

INDUSTRY INTERNSHIP (BINT803B)

2022 BATCH

Academic Year 2025-26

Name of the Student	:	**Samitha Halli**
USN	:	**1ST22AD043**
Name of the Internal guide	:	**Dr. Mamatha M**
Name of the External guide	:	**Mr. Manvith Shetty**
Details of the External guide	:	**Data Science Engineer**

INTERNSHIP DETAILS

Nature of the Internship	:	**Industry Internship**
Type of Internship	:	**Paid**
In case of Paid Internship, specify the amount paid in Rs	:	**4,950**
Place of the Internship (Organization name with Address)	:	**Dyashin Technosoft Pvt Ltd #32/33/5B, 1st Floor, Annapurna Compound Block II, Kanakapura Road, J P Nagar, Bengaluru, Karnataka – 560078, India.**
Title of the Internship	:	**Data Analytics using Python.**

SAMBHRAM INSTITUTE OF TECHNOLOGY

M. S. Palya, Bengaluru-560097

DEPARTMENT OFARTIFICIAL INTELLIGENCE & DATA SCIENCE

CERTIFICATE

This is to Certify that the Internship entitled **"Data Analytics using Python"** carried out at **Dyashin Technosoft Pvt Ltd, Bengaluru, Karnataka** has been successfully presented at **Sambhram Institute of Technology** for **VTU Internship** by **SAMITHA HALLI** bearing the **USN:1ST22AD043** in partial fulfillment for the award of degree in **Bachelor of Engineering in Artificial Intelligence and Data Science** of *Visvesvaraya Technological University, Belagavi* during academic year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Internship report has been approved as it satisfies the academic requirements in respect of internship work for the said degree.

Signature of Internal Guide	Signature of HOD	Signature of Principal
[Dr. Mamatha M] **Associate Professor,** **Dept of AI & DS, SaIT** **Bengaluru**	**[Dr. Nagaraja P]** **HOD,** **Dept of AI & DS, SaIT** **Bengaluru**	**[Dr. HG Chandrakanth]** **Principal,** **SaIT,** **Bengaluru**

EXTERNAL VIVA

Name of the Examiners	Signature with Date
1.	
2.	

SAMBHRAM INSTITUTE OF TECHNOLOGY

M. S. Palya, Bengaluru-560097

DEPARTMENT OFARTIFICIAL INTELLIGENCE & DATA SCIENCE

DECLARATION

I, **SAMITHA HALLI** bearing the USN:**1ST22AD043** student of **8th** semester B.E, Department of Artificial Intelligence and Data Science, Sambhram Institute of Technology, M. S. Palya, Bengaluru, hereby declare that the Internship entitled as the "**Data Analytics using Python.**" has been carried out under the external supervision of **Mr. Manvith Shetty, Data Science Engineer**, **Dyashin Technosoft Pvt Ltd, Bengaluru, Karnataka** and Internal supervision of **Dr. Mamatha M, Associate Professor, Department of AI&DS,** submitted in partial fulfillment of the course requirements for the award of degree in Bachelor of Engineering in Artificial Intelligence and Data Science of Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any other degree or diploma.

PLACE: BENGALURU
DATE:

NAME: SAMITHA HALLI
USN: 1ST22AD043

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crowned the efforts with success.

I would like to profoundly thank **Management of Sambhram Institute of Technology** for providing such a healthy environment for the successful completion of the Internship work.

I would like to express my thanks to **Dr. H.G. Chandrakanth**, Principal, **Sambhram Institute of Technology,** Bengaluru, for his encouragement that motivated me for the successful completion of Internship work.

It gives me immense pleasure to thank **Dr. Nagaraja P,** Professor and Head of Department of Artificial Intelligence and Data Science, for his constant support and encouragement.

I would like to express my deepest gratitude to my Internal Guide **Dr. Mamatha M**, Professor, Department of Artificial Intelligence and Data Science for her constant support and guidance throughout the Internship work.

I would also like to sincerely thank the Internship Coordinator **Dr. Mamatha M**, Associate Professor, for the valuable support, coordination, and assistance provided during the internship period, which helped in the successful completion of the internship.

I would like to express my deepest sense of gratitude to my External Guide **Mr. Manvith Shetty,** Data Science Engineer for his invaluable guidance and support.

I would like to express my deepest sense of gratitude to **Ms**. **Jashmi Pare,** branch head Dyashin Technosoft Pvt Ltd, Bengaluru

NAME: SAMITHA HALLI

USN: 1ST22AD043

ABSTRACT

The internship conducted at Dyashin Technosoft Pvt Ltd provided practical exposure to modern data analytics technologies and real-world business intelligence workflows. The internship was carried out under the domain of Data Analytics using Python, with a primary focus on data extraction, transformation, visualization, and dashboard development using industry-standard tools.

Throughout the internship period, various data analytics concepts were learned and implemented through practical tasks, mini projects, and collaborative activities. Core technologies including Python, SQL, Power BI, and Tableau were used to analyze datasets, derive meaningful insights, and create interactive visualizations. Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were extensively utilized for data manipulation and exploratory data analysis.

The internship additionally provided exposure to database querying using SQL, data modeling concepts, ETL processes, and business reporting methodologies. Practical experience was gained in connecting data sources, cleaning and transforming raw data, performing statistical analysis, and building comprehensive dashboards for business decision-making.

A significant portion of the internship involved contribution towards a real-time project focused on E-commerce Sales Analysis. This project encompassed data cleaning, sales trend identification, customer segmentation, revenue analysis, and the creation of interactive dashboards using Power BI and Tableau. The analysis helped identify key performance indicators, seasonal patterns, and actionable business insights from transactional data.

Dyashin

Innovate • Collaborate • Accelerate

Dear Samitha Halli,

We are pleased to extend an offer of internship with **Dyashin Technosoft Pvt. Ltd.** We believe that this experience will provide you with valuable insights and learning opportunities to further your career in your chosen field.

Position : Intern

Mentor Name : Manvith C S

Start Date : Monday, 02 February 2026

End Date : Monday, 01 June 2026

During your internship, you will undergo comprehensive learning in relevant technologies and will actively participate as a shadow resource in ongoing live projects. This hands-on experience will involve developing projects under the guidance and mentorship of our experienced team. The intern will be an integral part of daily stand-up meetings, where they will collaborate with project teams, share updates, and gain insight into project management processes. In addition, the intern will be responsible for sending daily status emails to their supervisors, ensuring clear communication and project alignment. This immersive experience will provide a well-rounded understanding of real-world project execution and foster professional growth.

We look forward to your contribution and the opportunity to work with you. If you accept this internship offer, please respond to this email by confirming your acceptance. You may reach out to HR personnel for any clarifications regarding the internship.

We believe this internship will be a mutually beneficial experience. Thank you for choosing **Dyashin Technosoft Pvt. Ltd.** as a part of your educational and career journey. We wish you all the very best.

Sincerely,

Jasmine

DYASHIN TECHNOSOFT PRIVATE LIMITED Bangalore

Authorised Signatory

www.dyashin.com | info@dyashin.com | +91 96069 58243

Dyashin

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Internship Completion Certificate

Date : 18 May 2026

Sr. No. : IN-BG-VCIP-0526-A849

To Whom It May Concern

This is to certify that **Samitha Halli (1ST22AD043)** , a student of **Sambhram Institute of Technology, Bengaluru**, has successfully completed an internship for a period of **02 February 2026 to 18 May 2026** at **Dyashin Technosoft Pvt Ltd., Bengaluru**.

During the internship, she worked extensively on **Python and Data Analytics** and contributed to various projects and tasks.

Samitha Halli, demonstrated exceptional skills in Python and Data Analytics and showed a keen interest in learning and applying new technologies. Her performance was commendable, and successfully met all the internship requirements and objectives. She also participated in a practical application case study to implement the learned concepts.

We wish her all the best in future endeavors.

Sincerely,

Jasmine

DYASHIN TECHNOSOFT PRIVATE LIMITED ★ Bangalore

Authorised Signatory

www.dyashin.com | info@dyashin.com | +91 96069 58243

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CHAPTER 1

INTRODUCTION

Internships serve as a critical bridge between theoretical academic knowledge and practical industrial experience. They enable students to understand professional work environments, enhance technical proficiency, and gain exposure to tools and methodologies currently employed in the industry. In the rapidly evolving field of Data Science and Analytics, internships provide invaluable opportunities to work with real datasets, modern visualization platforms, and analytical workflows that prepare students for industry demands.

The internship was carried out at Dyashin Technosoft Pvt Ltd under the domain of Data Analytics using Python. The primary objective was to gain practical knowledge in extracting insights from data using programming and visualization tools including Python, SQL, Power BI, and Tableau. The initial phase of the internship included orientation sessions that introduced the organization structure, professional expectations, project workflows, and current industry practices followed in analytics teams.

As part of the internship, various development and analytics tools including Python (with libraries such as Pandas, NumPy, Matplotlib, and Seaborn), SQL databases, Power BI Desktop, and Tableau Public were installed and configured for project implementation. The learning process was structured to progress gradually through data handling, exploratory analysis, statistical techniques, visualization development, and dashboard creation.

Data analytics activities involved learning Python fundamentals, data manipulation using Pandas, numerical operations with NumPy, and creating visualizations using Matplotlib and Seaborn. SQL was used for querying databases, extracting relevant data, performing aggregations, and preparing datasets for analysis. Power BI and Tableau were employed to build interactive dashboards that presented business insights in an accessible format.

The internship also provided exposure to data cleaning methodologies, ETL processes, business intelligence reporting, and collaborative project workflows. Participation in project discussions, review meetings, and team-based analysis activities contributed towards improving analytical thinking, communication skills, and professional discipline.

A major component of the internship involved contributing to a real-time project focused on E-commerce Sales Analysis. This project required analyzing transactional data to identify sales trends, customer behavior patterns, product performance, and revenue metrics. Interactive dashboards were developed to visualize key performance indicators and support data-driven business decisions.

The internship experience facilitated understanding of the complete data analytics lifecycle, including data collection, cleaning, transformation, analysis, visualization, and reporting. Practical exposure gained through the project implementation significantly improved confidence in handling real-world analytics tasks and strengthened overall career readiness in the field of data analytics and business intelligence.

The analytics industry continues to expand rapidly with organizations increasingly relying on data-driven insights for strategic decisions. Companies expect graduates to possess not only theoretical understanding but also practical implementation experience with industry-standard tools. Data analytics has emerged as one of the most sought-after domains because it enables organizations to transform raw data into actionable intelligence. Learning analytics technologies helps in understanding how data flows through systems and how insights are derived to drive business outcomes.

During the internship, emphasis was also placed on understanding collaborative analytics workflows and reporting methodologies followed in professional environments. Exposure to version control, project reviews, debugging practices, and task management activities helped in understanding how analytics teams coordinate activities efficiently. Continuous interaction with mentors and team members provided opportunities to improve technical discussion skills, analysis planning, and workflow management capabilities.

Beyond technical learning, the internship contributed towards improving adaptability, self-learning ability, and professional confidence. Real-time implementation activities encouraged independent problem-solving and analytical thinking while handling data quality issues, visualization challenges, and reporting requirements. This practical exposure played a significant role in strengthening industry readiness and understanding professional analytics standards.

In addition to technical learning, the internship contributed towards improving adaptability, self-learning ability, and professional confidence. Real-time implementation activities encouraged independent problem-solving and analytical thinking while handling frontend rendering issues, backend logic implementation, API communication, and deployment-related challenges. This practical exposure played an important role in strengthening industry readiness and understanding professional software engineering standards.

CHAPTER 2

ABOUT THE ORGANIZATION

2.1 Introduction to the Organization

Dyashin Technosoft Pvt Ltd is a technology solutions company established with the aim of delivering innovative software development, data analytics, and digital transformation services to clients across various industries. The organization focuses on leveraging modern technologies to solve complex business challenges and enable data-driven decision-making. With expertise spanning software development, business intelligence, and analytics consulting, Dyashin Technosoft serves as a technology partner for organizations seeking to harness the power of data.

2.2 History and Background

Dyashin Technosoft Pvt Ltd was established with the vision of becoming a trusted technology partner for businesses seeking digital transformation and analytics solutions. The organization was founded to address the growing demand for data-driven insights and modern software solutions in the market.

Over time, the company expanded its service offerings from software development to include comprehensive data analytics, business intelligence, visualization services, and technical training programs. The organization adopted modern technologies and collaborative workflows to deliver scalable and efficient solutions for clients ranging from startups to established enterprises.

Dyashin Technosoft has gradually evolved into a company that combines technical expertise with domain knowledge to create impactful solutions. The organization also focuses on talent development through structured internship programs that prepare students for careers in technology and analytics.

2.3 Vision and Mission

Vision

The vision of Dyashin Technosoft Pvt Ltd is to become a leading technology solutions provider that empowers organizations through innovative software and analytics services. The organization aims to create value for clients by transforming data into strategic assets and enabling informed decision-making across business functions.

through mentorship, referrals, collaboration, and verified professional networks irrespective of geographical limitations.

Mission

The mission of the organization is to deliver high-quality technology solutions that address real business challenges while fostering a culture of continuous learning and innovation. The organization strives to build long-term partnerships with clients by providing reliable, scalable, and cost-effective solutions.

The organization also focuses on developing talent through training programs and internships, contributing to the growth of skilled professionals in the technology industry.

2.4 Organizational Structure

Dyashin Technosoft Pvt Ltd follows a structured organizational hierarchy consisting of multiple departments responsible for different operational and technical functions. Each department collaborates to ensure efficient project execution, service delivery, and client satisfaction.

The organization includes a management team responsible for strategic planning, business development, and operational oversight. The technology and development teams handle software development, data analytics, visualization, system integration, and technical support activities.

The data analytics and business intelligence teams focus on data processing, analysis, dashboard development, reporting, and client consulting. Project management teams monitor development workflows, coordinate tasks, conduct review meetings, and ensure timely delivery of project milestones.

The organization also includes training and mentoring teams that provide technical guidance and learning support to interns and junior team members. Client services and support teams handle communication, requirement gathering, and ongoing support for delivered solutions.

2.5 Products and Services

Dyashin Technosoft Pvt Ltd provides a range of technology and analytics services designed to support business growth and operational efficiency.

Major services provided by the organization include:

- Custom software development and application maintenance
- Data analytics and business intelligence solutions
- Dashboard development using Power BI, Tableau, and similar platforms
- Database design, management, and optimization
- ETL development and data integration services
- Technical consulting and digital transformation advisory
- Training programs and internship opportunities
- Cloud solutions and deployment services

The organization also offers specialized services including predictive analytics, data visualization, reporting automation, and custom analytics solution development tailored to specific industry requirements.

Additional services include project monitoring, resume evaluation, ATS-based scoring systems, GitHub profile verification, and collaborative development support for technical projects.

2.6 Departments and Functions

Different departments within the organization perform specialized functions to maintain operational efficiency and deliver successful project outcomes.

Software Development Department

This department handles the design, development, testing, deployment, and maintenance of software applications. It manages frontend and backend development, API creation, system integration, and application support activities.

Data Analytics and BI Department

The analytics department focuses on data extraction, transformation, analysis, visualization, and reporting. It handles Power BI and Tableau dashboard development, SQL-based data querying, Python-based analysis, and business intelligence consulting.

Project Management Department

This department oversees project planning, resource allocation, timeline management, review meetings, and delivery coordination. It ensures smooth collaboration between teams and monitors project progress through established methodologies.

Training and Development Department

The training department conducts technical learning sessions, mentoring programs, and skill development activities for interns and employees. It supports career preparation and technical competency building.

Client Services Department

This department manages client communication, requirement gathering, solution delivery, and ongoing support. It works towards maintaining strong client relationships and ensuring satisfaction with delivered solutions.

2.7 Technologies and Tools Used

Dyashin Technosoft Pvt Ltd utilizes various modern technologies, frameworks, and tools to support software development, data analytics, and visualization activities.

Programming Languages

- Python
- SQL

Data Analytics Libraries

- Pandas
- NumPy

- Matplotlib
- Seaborn

Visualization Tools

- Power BI
- Tableau
- Excel

Database Technologies

- MySQL
- PostgerSQL
- Microsoft SQL Server
- No SQL

Development Tools

- Visual Studio Code
- Jupyter Notebook
- Power BI
- Anaconda

These technologies are extensively used for data processing, analysis, visualization, reporting, software development, and project delivery activities within the organization.

Several technologies and tools were used during the internship to perform data analysis, visualization, and reporting activities. Python was used for data cleaning, transformation, and analytical operations, while SQL was used for database querying and data management tasks.

Power BI and Tableau were used for creating interactive dashboards, charts, KPIs, and business intelligence reports. Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were utilized for data manipulation, statistical analysis, and graphical visualization.

CHAPTER 3

OBJECTIVES OF THE INTERNSHIP

The internship at Dyashin Technosoft Pvt Ltd was primarily focused on gaining practical exposure to data analytics technologies and understanding real-world workflows followed in analytics teams. The internship provided an opportunity to strengthen technical skills, build proficiency with industry tools, and apply theoretical concepts learned during academics to practical implementation activities.

The internship involved continuous learning, implementation tasks, visualization projects, collaborative discussions, and major project development activities related to Data Analytics using Python. Through practical exposure and project-based learning, a comprehensive understanding of data manipulation, statistical analysis, database querying, and dashboard development was achieved.

The internship also aimed to improve analytical thinking, problem-solving abilities, teamwork, communication skills, and understanding of professional analytics environments. Participation in review meetings, implementation discussions, testing activities, and project coordination sessions helped in gaining valuable industrial exposure.

3.1 Objectives of the Internship

3.1.1 To Gain Practical Exposure to Data Analytics

One of the primary objectives was to gain hands-on experience in data analytics using industry-standard tools and technologies. The internship provided practical exposure to the complete analytics workflow including data collection, cleaning, transformation, analysis, visualization, and reporting.

One of the primary objectives was to gain hands-on experience in data analytics using industry-standard tools and technologies. The internship provided practical exposure to the complete analytics workflow including data collection, cleaning, transformation, analysis, visualization, and reporting.

3.1.2 To Develop Python Programming Skills For Data Analysis

A key objective was to strengthen Python programming skills specifically for data analytics applications. Practical tasks involved using Pandas for data manipulation, NumPy for numerical computations, and Matplotlib and Seaborn for creating visualizations.

designing responsive layouts, implementing navigation systems, managing dynamic rendering, and creating interactive user interfaces.

Through assignments and project work, proficiency was developed in handling dataframes, cleaning datasets, performing aggregations, conducting statistical analysis, and generating visual representations of data patterns.

3.1.3 To Build Database Querying Skills using SQL

The internship focused on developing strong SQL skills for extracting and manipulating data from relational databases. Practical activities included writing queries for data selection, filtering, joining tables, aggregation, and creating derived datasets for analysis.

The internship focused on developing strong SQL skills for extracting and manipulating data from relational databases. Practical activities included writing queries for data selection, filtering, joining tables, aggregation, and creating derived datasets for analysis.

3.1.4 To Learn Data Visulaization Using Power BI and Tableau

Another important objective was to gain proficiency in business intelligence tools including Power BI and Tableau. The internship involved creating interactive dashboards, designing visualizations, implementing filters and slicers, and developing reports that communicate insights effectively. Learning these tools provided understanding of how organizations present data to stakeholders and support decision-making through visual analytics.

3.1.5 To Understand Real-Time Analytics Workflows

Understanding professional analytics workflows and methodologies was a major objective. The internship provided exposure to data governance practices, documentation standards, collaborative workflows, and project review processes followed in analytics teams.

One of the major objectives of the internship was to understand real-time analytics workflows and business intelligence practices followed in professional organizations. The internship provided practical exposure to data collection, data cleaning, transformation, visualization, dashboard creation, and report generation processes using Python, SQL, Power BI, and Tableau.

The internship also provided practical knowledge related to KPI tracking, sales trend analysis, customer behavior analysis, and business reporting techniques used in modern data-driven environments. These activities helped in understanding end-to-end analytics workflows followed in real-world business intelligence projects.

Participation in review discussions and project meetings helped in understanding how analytics projects are planned, executed, monitored, and delivered in organizational environments.

3.1.6 To Improve Problem-Solving and Analytical Skills

The internship aimed to improve logical thinking, troubleshooting capabilities, and analytical reasoning through continuous practice and project implementation.

During project work, various data quality issues, visualization challenges, query optimization problems, and reporting requirements were addressed through systematic analysis and problem-solving approaches. This helped in developing the ability to approach analytical challenges methodically.

3.1.7 To Enhance Communication and Teamwork Skills

The internship focused on improving communication abilities and collaboration skills through team-based project activities, review meetings, technical discussions, and presentation sessions.

Working with team members during project implementation helped in understanding task coordination, workflow planning, and collective problem-solving practices essential for professional environments.

3.1.8 To Apply Academic Knowledge in Practical Projects

A major objective was to apply academic concepts and theoretical knowledge in real-world analytics activities.

Concepts related to statistics, database management, programming fundamentals, and data structures studied during coursework were implemented through assignments and project development activities, bridging the gap between classroom learning and industrial practice.

One of the major objectives of the internship was to apply academic concepts and theoretical knowledge in real-time Data Analytics activities and project implementation tasks. Programming concepts, database management techniques, statistical methods, and visualization concepts learned during academics were practically implemented using Python, SQL, Power BI, and Tableau.

The internship provided opportunities to apply classroom knowledge through the E-Commerce Sales Analysis project, where data cleaning, transformation, analysis, and dashboard creation activities were carried out systematically. Practical implementation of analytical concepts helped in improving technical understanding and problem-solving abilities significantly.

This practical implementation helped in connecting classroom learning with industrial software engineering practices and improved conceptual understanding significantly.

3.1.9 To Gain Industry Exposure and Career Readiness

The internship aimed to improve industry readiness by providing exposure to professional analytics practices, tools, workflows, and organizational expectations.

Interaction with mentors, participation in review meetings, implementation activities, debugging sessions, and project evaluations helped in understanding current market requirements and software industry standards. The internship environment encouraged self-learning, adaptability, professional discipline, and continuous technical improvement.

The internship aimed to improve industry readiness by providing exposure to professional analytics practices, tools, workflows, and organizational expectations.

3.2 Scope of Work

The scope of work during the internship included Python-based data analysis, SQL database querying, Power BI dashboard development, Tableau visualization, data cleaning, exploratory data analysis, and reporting activities.

The internship involved implementation activities related to E-commerce Sales Analysis including sales trend identification, customer segmentation, product performance evaluation, revenue analysis, and interactive dashboard creation for business intelligence purposes.

The internship also included participation in analytical review meetings, project planning discussions, cleaning sessions, group activities, and implementation monitoring processes.

3.3 Duration and Location of Internship

The internship involved implementation activities related to E-commerce Sales Analysis including sales trend identification, customer segmentation, product performance evaluation, revenue analysis, and interactive dashboard creation for business intelligence purposes.

The internship was conducted in hybrid mode, involving both online learning and on-site project activities. This mode provided flexibility while enabling continuous interaction with mentors, team members, and project review teams.

CHAPTER 4

LEARNING EXPERIENCE

The internship at Dyashin Technosoft Pvt Ltd provided comprehensive practical exposure to data analytics practices, visualization techniques, and real-world workflows followed in analytics teams. Throughout the internship period, continuous learning activities, implementation tasks, visualization development, and project work were conducted to strengthen both technical and professional competencies.

The internship focused on Data Analytics using Python along with SQL, Power BI, and Tableau. The learning experience encompassed data manipulation, statistical analysis, database querying, visualization design, dashboard development, and reporting practices. Exposure to real-time implementation activities helped in understanding how organizations leverage data for strategic insights.

The internship also provided opportunities to work in team-based development environments where collaborative coding, project discussions, workflow planning, debugging sessions, and review meetings were conducted regularly. Participation in these activities improved understanding of industrial software development methodologies and professional communication practices.

4.1 Department Assigned

During the internship, work was carried out under the Data Analytics and Business Intelligence Department of Dyashin Technosoft Pvt Ltd. The department focused on delivering analytics solutions, visualization services, and business intelligence reporting for various clients.

The department handled data extraction, transformation, analysis, dashboard creation, and reporting activities. Work was carried out through collaborative practices, mentorship, and project-based learning methodologies.

Exposure to real-time development workflows helped in understanding how Power BI, Python services, databases, and deployment systems work together in modern web applications.

4.2 Learning and Orientation Activities

The internship began with orientation sessions focused on understanding the organization structure, internship workflow, professional conduct, and industry practices followed in analytics teams.

Initial learning activities included installation and configuration of tools such as Python (Anaconda distribution), Jupyter Notebook, Visual Studio Code, MySQL, Power BI Desktop, and Tableau Public. These setup activities helped in understanding development environments required for analytics work.

The internship also included sessions related to communication standards, project review methods, collaborative workflows, attendance management, project submission guidelines, and industrial professionalism. These sessions helped in understanding workplace discipline, responsibility management, and teamwork practices followed in organizational environments.

4.3 Daily and Weekly Activities

The internship involved continuous daily and weekly activities related to learning, project development, review discussions, and Dashboard Creation, analyzing Data. Each week focused on specific technologies and implementation concepts, gradually progressing from foundational learning to advanced project execution activities.

4.3.1 Python Programming Activities

A significant portion of the internship focused on Python programming for data analytics.

Initial activities involved understanding Python fundamentals including variables, data types, operators, control structures, functions, and file handling. Practical exercises were conducted to build programming proficiency.

Data manipulation activities focused on Pandas library for loading datasets, handling dataframes, filtering data, performing groupby operations, merging datasets, and handling missing values. NumPy was used for numerical computations, array operations, and statistical calculations.

Visualization activities involved creating plots using Matplotlib and Seaborn including line charts, bar graphs, histograms, scatter plots, heatmaps, and distribution plots. These visualizations helped in exploring data patterns and presenting findings effectively.

React.js learning activities involved component creation, JSX implementation, props handling, hooks, conditional rendering, list rendering, routing, state management, and API integration. Frontend interfaces were integrated with backend services to manage dynamic data rendering and application workflows.

4.3.2 SQL and Database Activities

SQL learning activities included understanding database concepts, writing queries, and extracting data for analysis.

Practical activities involved SELECT statements, WHERE clauses, JOIN operations, GROUP BY aggregations, subqueries, and data filtering techniques. Queries were written to extract, transform, and prepare data for analysis and visualization.

Database connectivity was established between Python and SQL databases to enable data extraction and integration into analysis workflows.

Backend project structures were organized systematically to improve scalability, maintainability, and workflow execution during project development.

4.3.3 Power BI Development Activities

Power BI learning included data import, transformation using Power Query, data modeling, DAX calculations, and visualization development.

Activities involved creating various charts, tables, cards, and maps to represent data visually. Interactive features including slicers, filters, drill-through capabilities, and bookmarks were implemented to enable dynamic exploration of data.

Dashboard development focused on designing layouts that present key metrics and insights in an accessible format for business users.

4.3.4 Tableau Visualization Activities

Tableau learning activities included connecting data sources, creating worksheets, building visualizations, and assembling dashboards.

Practical activities involved creating calculated fields, implementing filters, building hierarchies, and designing interactive visualizations. Story features were explored to create narrative presentations of analytical findings.

Tableau Public was used to publish and share completed visualizations.

4.3.5 Data Cleaning and Transformation Activities

Data cleaning activities were conducted throughout the internship to prepare datasets for analysis. Tasks included handling missing values, removing duplicates, correcting data types, standardizing formats, handling outliers, and transforming data into suitable structures for analysis and visualization.

ETL concepts were applied to extract data from sources, transform it according to requirements, and load it into analytical tools for processing.

4.3.6 Review Meetings and Team Discussions

Regular review meetings were conducted throughout the internship period.

Meetings were used to discuss completed tasks, implementation progress, challenges encountered, and next steps. Participation in these meetings helped in understanding project monitoring and collaborative practices followed in professional environments.

Regular review meetings and team discussions were conducted during the internship to monitor the progress of the project activities. These sessions helped in understanding project requirements, resolving technical issues, and improving analytical approaches.

4.4 Roles and Responsibilities

During the internship, multiple technical and analytical responsibilities were handled under the role of Data Analytics Intern at Dyashin Technosoft Pvt Ltd.

4.4.1 Data Analysis Responsibilities

Data analysis responsibilities included collecting, cleaning, transforming, and analyzing datasets using Python and SQL. Tasks involved handling missing values, organizing datasets, performing data filtering, executing SQL queries, and generating meaningful analytical insights from the E-Commerce sales dataset. Additional responsibilities included identifying sales trends, customer behavior patterns, and profit distribution for business analysis purposes.

4.4.2 Dashboard Development Responsibilities

Dashboard development responsibilities included creating interactive dashboards and business intelligence reports using Power BI and Tableau. Tasks involved designing charts, KPIs, filters, graphs, and visualization components to represent sales performance and business insights effectively. Responsibilities also included improving dashboard readability, maintaining visualization consistency, and generating analytical reports for decision-making purposes.

4.4.3 Project Collaboration Responsibilities

Team collaboration formed an important part of the internship responsibilities. Participation in review meetings, project discussions, implementation planning sessions, and technical activities helped in understanding collaborative workflows and professional analytical practices. Responsibilities included monitoring project progress, discussing dashboard improvements, analyzing workflows, and coordinating with teammates during project implementation activities.

Participation in review meetings, project discussions, implementation planning sessions, and analytical activities helped in understanding collaborative data analytics workflows and professional industry practices. Responsibilities included monitoring project progress, discussing dashboard improvements, analyzing datasets, and coordinating with teammates during the E-Commerce Sales Analysis project. Collaborative activities improved understanding of workflow management, project coordination methods, teamwork, and professional communication practices followed in Data Analytics environments.

4.4.4 Documentation and Reporting Responsibilities

Responsibilities also included preparing project documentation, generating analytical reports, validating dataset accuracy, and maintaining proper workflow records throughout the internship period. Reporting activities involved summarizing business insights, organizing project findings, and presenting analytical information through dashboards and visualization techniques. Continuous monitoring and report preparation activities improved understanding of professional documentation and reporting standards followed in Data Analytics environments.

4.5 Tasks Completed During the Internship

Several practical analytical and reporting tasks were completed throughout the internship period to strengthen technical knowledge and implementation capabilities in the field of Data Analytics and Business Intelligence.

4.5.1 Environment Setup Tasks

Initial tasks involved installing and configuring analytical and development tools including Python, Jupyter Notebook, Visual Studio Code, MySQL, Power BI, Tableau, and Microsoft Excel. Dataset management activities, software configuration, library installation, and workspace setup procedures were also performed during the initial phase of the internship.

4.5.2 Data Analysis and Visualization Tasks

Data analysis tasks included collecting datasets, performing data cleaning, handling missing values, transforming data, and executing SQL queries for analytical purposes. Visualization tasks involved creating charts, graphs, KPIs, dashboards, and analytical reports using Power BI and Tableau. Interactive dashboards were developed to represent sales trends, profit analysis, customer behavior, and regional performance insights effectively.

4.5.3 Database and Reporting Tasks

Database-related tasks involved managing structured datasets, executing SQL operations, filtering records, performing joins, and generating summarized business reports. Analytical reporting activities included preparing sales reports, monitoring performance indicators, and organizing data-driven

4.5.4 Testing and Debugging Tasks

Testing and debugging tasks involved validating dataset accuracy, identifying inconsistencies, resolving formatting issues, and improving dashboard performance. Additional activities included correcting visualization errors, validating SQL query outputs, monitoring report generation workflows, and ensuring proper functioning of analytical dashboards and reporting systems throughout the project implementation process.

4.6 Projects Handled During the Internship

The internship included implementation of analytical activities and one major project related to Data Analytics, Business Intelligence, and real-world business reporting practices.

4.6.1 Data Visualization and Dashboard Activities

During the internship, multiple dashboard creation and data visualization activities were carried out using Power BI and Tableau. These activities focused on creating interactive charts, KPIs, graphs, filters, and business reports for effective analysis and presentation of sales data. Visualization tasks helped in understanding dashboard design principles, report generation workflows, and business intelligence methodologies followed in modern organizations.

The activities also involved representing customer behavior, sales trends, regional performance, and profit distribution through analytical dashboards. Continuous practice in visualization improved understanding of graphical reporting techniques and interactive business reporting systems.

4.6.2 Data Analysis and SQL Activities

Data analysis activities included data cleaning, transformation, dataset organization, SQL query execution, and analytical report preparation using Python and SQL. Tasks involved handling missing values, filtering records, performing joins, sorting datasets, and generating summarized analytical insights for business reporting purposes.

These activities improved understanding of structured data management, database operations, analytical workflows, and real-time reporting methodologies used in professional Data Analytics environments.

4.6.3 E-Commerce Sales Analysis Dashboard (Major Project)

The major project handled during the internship was an E-Commerce Sales Analysis Dashboard developed using Python, SQL, Power BI, and Tableau. The project focused on analyzing sales datasets to identify customer purchasing behavior, sales performance, regional trends, and profit distribution candidate profile assessment. Frontend interfaces, backend APIs, and database systems were integrated to manage candidate evaluation workflows effectively.

Major project activities included data collection, data cleaning, transformation, SQL query execution, dashboard development, KPI implementation, report generation, and visualization of business insights through interactive dashboards. Various charts, filters, and graphical reports were created to improve analytical understanding and support business decision-making processes.

The project provided practical exposure to real-time Data Analytics workflows, business intelligence methodologies, dashboard development practices, and professional reporting standards followed in modern organizations. Continuous implementation and project review activities significantly improved technical knowledge, analytical thinking, and practical understanding of industrial data analytics practices.

4.7 Software, Tools, and Technologies Used

Several modern analytical tools, software platforms, and technologies were used throughout the internship period for data analysis, visualization, reporting, and business intelligence activities.

4.7.1 Programming Languages

- Python
- SQL

4.7.2 Data Visualization and Business Intelligence Tools

- Power BI
- Tableau
- Microsoft Excel

4.7.3 Python Libraries

- Pandas
- NumPy

- Matplotlib
- Seaborn

4.7.4 Development and Analytical Tools

- Jupyter Notebook
- Visual Studio Code
- MySQL
- Google Colab

4.7.5 Other Technologies and Concepts

- Data Cleaning
- Data Transformation
- Dashboard Development
- KPI Analysis
- Business Intelligence Reporting
- Data Visualization

These technologies and tools were extensively used for data cleaning, transformation, visualization, dashboard creation, SQL query execution, analytical reporting, and business intelligence workflow implementation activities throughout the internship period.

4.8 Technical Contributions

Several technical contributions were made throughout the internship through data analysis, dashboard development, visualization, reporting, testing, and workflow optimization activities.

4.8.1 Data Analysis Contributions

Data analysis contributions included performing data cleaning, handling missing values, transforming datasets, and executing SQL queries for business analysis purposes. Python libraries such as Pandas and NumPy were used to organize, process, and analyze e-commerce sales datasets efficiently. Analytical activities also included identifying sales trends, customer behavior patterns, and profit distribution insights for reporting purposes.

4.8.2 Dashboard and Visualization Contributions

Dashboard and visualization contributions involved creating interactive dashboards, charts, KPIs, and graphical reports using Power BI and Tableau. Visualization workflows were implemented to represent regional sales performance, customer purchasing behavior, revenue trends, and business insights effectively. Additional contributions included improving dashboard readability, maintaining visualization consistency, and optimizing report presentation techniques.

4.8.3 Database and Reporting Contributions

Database-related contributions included executing SQL operations, filtering records, performing joins, and generating summarized analytical reports from structured datasets. Reporting contributions involved preparing business intelligence reports, monitoring KPI performance, and organizing analytical findings for better understanding of business workflows and decision-making processes.

4.8.4 Testing and Optimization Contributions

Continuous testing and optimization activities were performed throughout the internship period. Contributions included validating dataset accuracy, resolving formatting inconsistencies, correcting visualization issues, improving dashboard performance, and ensuring proper execution of analytical workflows. Additional activities involved testing report outputs, optimizing data representation techniques, and maintaining stability and accuracy throughout the project implementation process.

CHAPTER 5

LEARNING OUTCOMES

The internship at Dyashin Technosoft Pvt Ltd provided valuable learning experiences that significantly improved technical knowledge, practical implementation abilities, professional communication skills, and understanding of industrial analytics practices. Continuous participation in analysis activities, visualization development, project work, and review meetings helped in gaining real-world exposure to data analytics environments.

The internship not only strengthened theoretical understanding but also provided opportunities to apply academic concepts through practical implementation in the E-commerce Sales Analysis project.

The internship helped in gaining practical knowledge of Python, SQL, Power BI, and Tableau for data analytics applications. It also improved skills in data visualization, dashboard development, teamwork, and real-time project handling.

The learning outcomes achieved during the internship are explained below.

5.1 Technical Knowledge Gained

One of the most significant outcomes was the improvement of technical knowledge related to data analytics and visualization.

Python Proficiency: The internship significantly enhanced Python programming skills for data analytics. Knowledge of Pandas for data manipulation, NumPy for numerical operations, and Matplotlib and Seaborn for visualization was developed through practical exercises and project implementation.

Understanding of data structures, dataframe operations, aggregations, merging, pivoting, and statistical analysis using Python was strengthened through continuous practice.

SQL Skills: SQL querying skills were developed through practical database activities. Knowledge of query syntax, joins, aggregations, subqueries, and data extraction techniques improved significantly.

Understanding of database structures, relationships, and query optimization helped in efficiently preparing data for analysis.

Power BI Proficiency: Proficiency in Power BI was gained including data import, Power Query transformations, data modeling, DAX calculations, and visualization development.

Tableau Skills: Knowledge of Tableau was developed including data connection, worksheet creation, visualization building, calculated fields, and dashboard assembly.

Understanding of interactive visualization features and publishing capabilities was gained through practical implementation.

Data Preparation: Knowledge of data cleaning, transformation, and preparation was enhanced. Understanding of handling missing values, duplicates, outliers, and data type conversions was developed through practical application.

SQL skills were strengthened through writing and executing queries on real datasets. Concepts including aggregation functions, subqueries, JOIN operations, conditional filtering, and data grouping were applied practically during the project.

Knowledge of Power BI and Tableau improved from basic familiarity to the ability to build complete, multi-page interactive dashboards. Skills in connecting data sources, applying transformations, creating calculated fields, designing visual layouts, and adding interactivity through filters and slicers were developed during the internship.

The internship provided substantial technical knowledge in data analytics tools and technologies. Proficiency in Python improved considerably through consistent practice with Pandas, NumPy, Matplotlib, and Seaborn. The ability to clean messy datasets, perform aggregations, filter and transform data, and generate insightful visualizations was developed through repeated hands-on tasks.

SQL skills were strengthened through writing and executing queries on real datasets. Concepts including aggregation functions, subqueries, JOIN operations, conditional filtering, and data grouping were applied practically during the project.

Python programming was extensively used for data cleaning, transformation, and analysis activities. Libraries such as Pandas, NumPy, Matplotlib, and Seaborn helped in understanding data manipulation, statistical analysis, and graphical representation techniques. SQL concepts including database querying, filtering, sorting, joins, and aggregation functions were also practiced to manage and analyze structured datasets efficiently.

5.2 Practical Exposure

The internship provided extensive practical exposure to real-world analytics environments.

Unlike theoretical learning, the internship involved direct hands-on implementation through assignments, exercises, and project development. Practical exposure began with environment setup and progressed through increasingly complex analysis and visualization tasks.

Working on the E-commerce Sales Analysis project provided experience in handling real transactional data, identifying business patterns, and creating actionable visualizations. The project involved the complete analytics workflow from data preparation to dashboard delivery.

Practical exposure was gained in connecting various data sources, transforming data for analysis, performing exploratory analysis, and developing visualizations that communicate findings effectively.

Understanding of how analytics supports business decision-making was developed through practical implementation of metrics, KPIs, and interactive reporting.

Unlike academic assignments with predefined outcomes, the internship involved working with actual business data that required judgment and problem-solving at every step. Practical exposure was gained in handling real-world data quality issues, understanding business context for analysis, and structuring analytical outputs for stakeholder consumption.

The end-to-end experience of working on the E-commerce Sales Analysis project — from raw data to a finished dashboard provided a clear understanding of how analytics projects are executed in professional environments.

Practical exposure was mainly gained through the E-Commerce Sales Analysis project, where sales datasets were analyzed to identify trends, customer behavior, profit distribution, and regional performance insights. Activities such as preparing datasets, handling missing values, generating reports, and designing interactive dashboards improved analytical thinking and technical implementation skills. These tasks helped in understanding real-world business analytics methodologies and reporting workflows used in professional environments.

The internship also provided opportunities to participate in review meetings, technical discussions, and collaborative activities related to project development and report generation.

5.3 Communication and Teamwork Skills

The internship improved communication abilities and collaboration skills through continuous interaction with mentors and team members.

Participation in project discussions, review meetings, and presentation sessions improved confidence in presenting technical findings and discussing analytical approaches.

Working collaboratively during project implementation helped in understanding task coordination, workflow planning, and collective problem-solving practices.

Regular interaction with mentors provided opportunities to clarify concepts, receive feedback, and improve analytical methodologies.

The ability to present analytical findings to non-technical stakeholders improved significantly during the internship. Participating in review meetings, discussing project progress, and explaining dashboard insights in simple terms helped build professional communication confidence. Report writing and documentation skills were also refined through the preparation of project summaries and analytical narratives.

The internship played an important role in improving communication abilities and teamwork skills through continuous interaction with mentors, teammates, and review teams. Participation in project discussions, review meetings, and presentation sessions helped in improving confidence in explaining technical concepts, analytical workflows, and project implementation activities professionally.

Working collaboratively during the E-Commerce Sales Analysis project helped in understanding task coordination, workflow planning, and collective problem-solving approaches. Team discussions related to dashboard development, data visualization, and report preparation improved the ability to share ideas, receive feedback, and coordinate effectively in a professional environment.

Regular interaction with mentors also improved professional communication practices, technical discussion abilities, and responsibility management skills. Overall, the internship enhanced interpersonal skills, teamwork capabilities, and professional discipline required for working efficiently in organizational environments.

5.4 Problem-Solving Abilities

The internship improved analytical thinking and problem-solving abilities.

During project work, various data quality issues, visualization challenges, and analytical problems were encountered and resolved through systematic approaches.

Experience in debugging data issues, optimizing queries, resolving visualization problems, and addressing reporting requirements strengthened troubleshooting capabilities.

The ability to analyze problems systematically, identify root causes, and implement appropriate solutions was developed through continuous practice.

Working with real data presented several challenges, including inconsistent data formats, missing values, ambiguous category labels, and performance issues in dashboards with large datasets. Resolving these issues independently and through team discussions strengthened analytical thinking and problem-solving abilities. Debugging Python scripts and optimizing SQL queries were also important aspects of this development.

The internship significantly improved analytical thinking and problem-solving abilities through continuous practical implementation and project activities. During the E-Commerce Sales Analysis project, several challenges related to data cleaning, missing values, incorrect formatting, and dashboard visualization were identified and resolved using Python, SQL, Power BI, and Tableau techniques.

Continuous practice in data analysis and visualization helped in improving the ability to analyze datasets systematically and generate meaningful business insights. Troubleshooting issues related to SQL queries, report generation, chart formatting, and data transformation activities strengthened logical thinking and technical problem-solving skills.

5.5 Industry Practices Learned

The internship provided exposure to professional analytics workflows and industry practices.

Documentation Standards: Understanding of documentation practices for analytics projects including maintaining records, preparing reports, and communicating methodologies was developed.

Review Processes: Exposure to project review methodologies helped in understanding how analytics work is monitored, evaluated, and improved through continuous feedback.

Data Governance: Understanding of data quality practices, validation techniques, and governance standards was gained through project implementation.

Reporting Practices: Knowledge of business reporting standards, dashboard design principles, and visualization best practices was developed through practical experience

The internship provided insight into how data analytics teams operate in a professional environment. Key practices learned include data governance and documentation, maintaining clean and reproducible analytical code, following a structured analytical workflow from problem definition to reporting, using version control for scripts and project files, and presenting findings in a format aligned with business decision-making needs.

The internship provided valuable exposure to industrial data analytics practices and professional workflows followed in modern organizations. Practical understanding was gained in areas such as data cleaning, report generation, dashboard development, data visualization, and business intelligence methodologies using Python, SQL, Power BI, and Tableau.

Participation in project review meetings, technical discussions, and collaborative activities helped in understanding workflow planning, task coordination, and project monitoring practices followed in professional environments. Exposure to real-time business datasets also improved understanding of how organizations use analytical tools for decision-making and performance analysis.

The internship further improved knowledge related to professional documentation, report preparation, data presentation techniques, and dashboard reporting standards. Overall, the internship helped in understanding industrial analytics workflows, teamwork practices, and professional standards commonly followed in the field of Data Analytics and Business Intelligence.

The internship provided valuable exposure to industrial data analytics practices and professional workflows followed in modern organizations. Practical understanding was gained in areas such as data cleaning, report generation, dashboard development, data visualization, and business intelligence methodologies using Python, SQL, Power BI, and Tableau.

Participation in project review meetings, technical discussions, and collaborative activities helped in understanding workflow planning, task coordination, and project monitoring practices followed in professional environments. Exposure to real-time business datasets also improved understanding of how organizations use analytical tools for decision-making and performance analysis.

5.6 Application of Academic Concepts

The internship provided opportunities to apply theoretical knowledge and academic concepts in practical Data Analytics activities. Concepts learned during academics such as programming fundamentals, database management, statistics, data visualization, and analytical techniques were implemented practically through project development and real-time data analysis tasks.

Python programming concepts including variables, functions, loops, conditional statements, and libraries were applied during data cleaning, transformation, and analysis activities. SQL concepts such as database querying, filtering, sorting, joins, and aggregation functions were used to manage and analyze sales datasets efficiently during the E-Commerce Sales Analysis project.

Academic knowledge related to data visualization and business intelligence was implemented using Power BI and Tableau for creating interactive dashboards, KPIs, charts, and analytical reports. These activities helped in understanding how theoretical concepts are applied in professional environments for business analysis and decision-making purposes.

The internship also improved understanding of report preparation, workflow management, and project documentation methodologies followed in organizational environments. Practical implementation of academic concepts significantly improved technical confidence, analytical thinking, and readiness for future career opportunities in the field of Data Analytics and Business Intelligence.

The internship provided opportunities to apply theoretical knowledge and academic concepts in practical Data Analytics activities. Concepts learned during academics such as programming fundamentals, database management, statistics, data visualization, and analytical techniques were implemented practically through project development and real-time data analysis tasks.

Academic knowledge related to data visualization and business intelligence was implemented using Power BI and Tableau for creating interactive dashboards, KPIs, charts, and analytical reports. These activities helped in understanding how theoretical concepts are applied in professional environments for business analysis and decision-making purposes.

Concepts related to statistics and data interpretation were also applied while analyzing customer behavior, sales performance, profit distribution, and regional sales trends. Graphical representation methods learned during academics were used for creating meaningful visualizations and business reports.

Knowledge related to Software Development Life Cycle (SDLC), project planning, documentation, and workflow management was practically experienced during project execution and review meetings. The internship also helped in understanding how academic problem-solving approaches are used to handle real-world analytical challenges effectively.

Practical implementation of academic concepts significantly improved technical confidence, analytical thinking, and readiness for future career opportunities in the field of Data Analytics and Business Intelligence.

Python programming concepts including variables, functions, loops, conditional statements, and libraries were applied during data cleaning, transformation, and analysis activities. SQL concepts such as database querying, filtering, sorting, joins, and aggregation functions were used to manage and analyze sales datasets efficiently during the E-Commerce Sales Analysis project.

Overall, the internship successfully transformed academic knowledge into practical implementation skills and improved readiness for professional software development careers in the field of Data analytics Using Python .

CONCLUSION

The internship at Dyashin Technosoft Pvt Ltd was a highly valuable and enriching learning experience that provided practical exposure to Data Analytics and Business Intelligence technologies used in real-world industrial environments. The internship successfully helped in bridging the gap between academic learning and practical implementation by providing opportunities to work on real-time analytical tasks, dashboard development activities, data visualization techniques, and business reporting methodologies.

Throughout the internship period, continuous learning sessions, practical assignments, technical discussions, project reviews, and implementation activities significantly improved both technical and professional competencies. Practical exposure to Python, SQL, Power BI, and Tableau strengthened understanding of data analysis workflows, business intelligence practices, and visualization methodologies followed in modern organizations. The internship also improved the ability to work with structured datasets, generate meaningful business insights, and create interactive analytical dashboards for decision-making purposes.

One of the major achievements during the internship was the successful completion of the E-Commerce Sales Analysis project. The project provided practical understanding of data collection, data cleaning, transformation, visualization, report generation, and dashboard development processes. Various analytical techniques were implemented to identify sales trends, customer purchasing behavior, regional performance, and profit distribution patterns. Interactive dashboards and visual reports were created using Power BI and Tableau to present business insights effectively and improve analytical understanding.

The internship also contributed significantly towards improving professional skills such as communication, teamwork, analytical thinking, problem-solving abilities, and technical documentation practices. Participation in review meetings, team discussions, and project implementation activities improved confidence in presenting ideas, discussing workflows, and handling project-related responsibilities in a professional manner. Exposure to collaborative environments helped in understanding industrial work culture, task coordination methods, and workflow management practices followed in organizations.

Another important outcome of the internship was the practical application of academic concepts in real-time analytical activities. Concepts related to programming, database management, statistics, visualization, and reporting learned during academics were implemented successfully during project execution. This practical exposure improved conceptual clarity and strengthened confidence in applying theoretical knowledge to solve real-world business problems using analytical tools and technologies.

The internship experience also improved understanding of professional reporting standards, business intelligence workflows, and data-driven decision-making methodologies. Exposure to real-world datasets and analytical processes enhanced technical confidence and improved readiness for future career opportunities in the field of Data Analytics and Business Intelligence.

Overall, the internship at Dyashin Technosoft Pvt Ltd was a successful and productive experience that contributed significantly towards technical growth, professional development, industry exposure, and career preparation. The knowledge and practical skills gained during the internship will be highly beneficial for future academic projects, higher studies, and professional opportunities in the domains of Data Analytics, Business Intelligence, and Information Technology.

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Weekly Diary Entries

Week 1:

- Attended the internship orientation session and received an overview of the organization, internship objectives, and expected deliverables.
- Understood the data analytics workflow followed at Dyashin Technosoft Pvt Ltd and the tools that would be used during the internship.
- Installed and configured the Python environment including Anaconda, Jupyter Notebook, and VS Code, along with necessary libraries such as Pandas, NumPy, Matplotlib, and Seaborn.
- Installed MySQL Workbench, Power BI Desktop, and Tableau Public for use in later weeks.
- Reviewed sample datasets provided by the mentor to understand the type of analytical work expected during the internship.

Week 2:

- Began Python training with a focus on data loading, inspection, and basic data operations using Pandas.
- Practiced reading CSV and Excel files, understanding DataFrame structure, checking data types, and reviewing summary statistics.
- Learned to handle missing values using methods such as fillna(), dropna(), and interpolation techniques.
- Practiced removing duplicate records and standardizing column names and data types.
- Completed exercises on filtering and slicing DataFrames based on conditions.

Week 3:

- Focused on data transformation and feature engineering using Pandas — creating new calculated columns, applying functions using apply(), and grouping data using groupby().
- Practiced data aggregation operations including sum, mean, count, min, and max on grouped data

- Learned to merge and concatenate DataFrames using merge() and concat() operations.
- Worked with date and time data using Pandas datetime functions to extract year, month, and day components.
- Submitted a mini-exercise on data cleaning and transformation to the mentor for review and feedback.

Week 4:

- Started data visualization training using Matplotlib and Seaborn.
- Created bar charts, pie charts, histograms, and line graphs to visualize distributions and trends in sample datasets.
- Used Seaborn to create heatmaps, box plots, and scatter plots for exploring relationships between variables.
- Practiced formatting charts by adding titles, axis labels, legends, and color palettes for better readability.
- Participated in a team discussion on visualization best practices and how charts should be designed for business audiences.

Week 5:

- Introduced to SQL and its role in data analytics workflows.
- Practiced basic SQL commands including SELECT, FROM, WHERE, ORDER BY, and LIMIT.
- Learned aggregate functions such as SUM(), AVG(), COUNT(), MIN(), and MAX() and applied them with GROUP BY clauses.
- Wrote queries to extract monthly sales totals, top categories by revenue, and order counts by region from a sample e-commerce database.
- Practiced INNER JOIN, LEFT JOIN, and CROSS JOIN operations to combine data from multiple related tables.

Week 6:

- Continued SQL practice with advanced queries including subqueries, HAVING clause filtering, and window functions such as ROW_NUMBER() and RANK().
- Wrote queries to calculate cumulative revenue, rank products by sales, and identify top-performing customers.
- Practiced exporting SQL query results to CSV for further analysis in Python.
- Participated in a review meeting with the mentor to discuss SQL progress and received feedback on query optimization.
- Explored the e-commerce dataset that would be used for the major project and documented the key fields and their significance.

Week 7:

- Began working with the E-commerce Sales dataset — the foundation of the major internship project.
- Performed an initial data audit to identify missing values, inconsistent entries, incorrect data types, and outliers.
- Cleaned the dataset using Python — handled nulls, corrected date formats, removed duplicates, and standardized categorical values.
- Created a cleaned version of the dataset and documented all transformations performed.
- Discussed the project scope and dashboard requirements with the mentor and finalized the key metrics to be analyzed..

Week 8:

- Conducted Exploratory Data Analysis (EDA) on the cleaned e-commerce dataset using Python.
- Analyzed total sales, total profit, and order volume trends across different time periods.
- Created visualizations for product category performance, sub-category revenue distribution, and regional sales comparison.
- Investigated the relationship between discount rates and profit margins using scatter plots and correlation analysis.
- Documented EDA findings in a structured report and shared initial insights with the mentor for review.

Week 9:

- Continued EDA — analyzed customer segmentation, shipping mode preferences, and delivery time patterns.
- Identified the top 10 products by revenue and the top 5 cities by total orders using Pandas

- Wrote SQL queries to validate Python-based findings and cross-check data summaries.
- Began creating a structured Excel summary with pivot-table style outputs for use in dashboards.
- Participated in a mid-project review meeting and presented initial analytical findings to the mentor and team.

Week 10:

- Started Power BI dashboard development — imported the cleaned e-commerce dataset and reviewed it in Power Query.
- Built data model relationships between order, product, customer, and regional data tables.
- Created the first set of visualizations: total sales KPI card, monthly sales trend line chart, and region-wise sales map.
- Added slicers for Category, Region, and Year to enable interactive filtering of the dashboard.
- Tested the dashboard for accuracy and verified all visual outputs against the Python analysis results.

Week 11:

- Expanded the Power BI dashboard with additional report pages covering product performance, profit analysis, and discount impact.
- Created DAX measures for calculated fields such as Profit Margin %, Year-over-Year Growth, and Average Order Value.
- Designed a sub-category performance bar chart and a top products ranked table.
- Refined the dashboard layout, applied consistent color themes, and improved visual formatting for professional presentation.
- Shared the Power BI dashboard with the mentor for feedback and incorporated suggested improvements.

Week 12:

- Began building the Tableau version of the E-commerce Sales Analysis Dashboard.
- Connected Tableau to the cleaned dataset and created individual worksheets for sales trend, regional map, category breakdown, and profit analysis.
- Used calculated fields and parameters in Tableau to add dynamic filtering capabilities to the dashboard.
- Assembled individual worksheets into a cohesive Tableau dashboard with navigation and visual consistency.
- Compared the Power BI and Tableau dashboards and noted the strengths and differences of each platform.

Week 13:

- Refined both the Power BI and Tableau dashboards based on mentor feedback received during the previous review session.
- Added additional insights to the dashboards including seasonal trends, month-over-month sales growth, and profit hotspots by region.
- Wrote SQL queries to generate final summary tables for monthly revenue, category-level profitability, and discount distribution, which were cross-checked against dashboard outputs.
- Began drafting the analytical report summarizing the project findings, methodology, and key insights.
- Participated in a team discussion on data storytelling and how to present complex insights to non-technical stakeholders.

Week 14:

- Finalized all visualizations in Power BI and Tableau and performed thorough testing of filters, slicers, and calculated fields.
- Completed the analytical report covering the problem statement, data description, cleaning methodology, EDA findings, dashboard overview, and business insights.
- Prepared a project presentation summarizing the objectives, approach, tools used, key findings, and visual outputs.
- Presented the completed E-commerce Sales Analysis Dashboard to the mentor and received positive feedback along with minor suggestions for improvement.
- Incorporated final changes and ensured all project files were organized and well-documented.

Week 15:

- Conducted final review and quality check of all project deliverables including cleaned datasets, Python scripts, SQL query files, Power BI report, Tableau dashboard, and analytical report.
- Participated in the internship closing discussion, reviewed overall learning outcomes, and received feedback on performance from the mentor.
- Submitted all project deliverables and final documentation to the mentor and the organization.
- Reflected on skills acquired during the internship and identified areas for continued learning in advanced analytics, machine learning, and cloud-based BI platforms.
- Expressed gratitude to the team at Dyashin Technosoft Pvt Ltd for the learning opportunity and mentorship provided throughout the internship period.